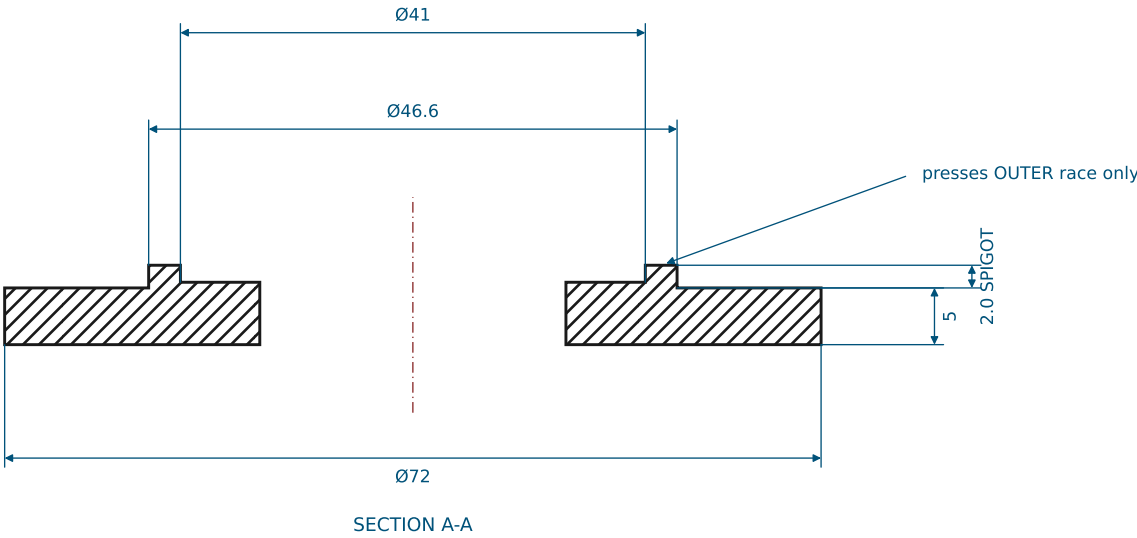
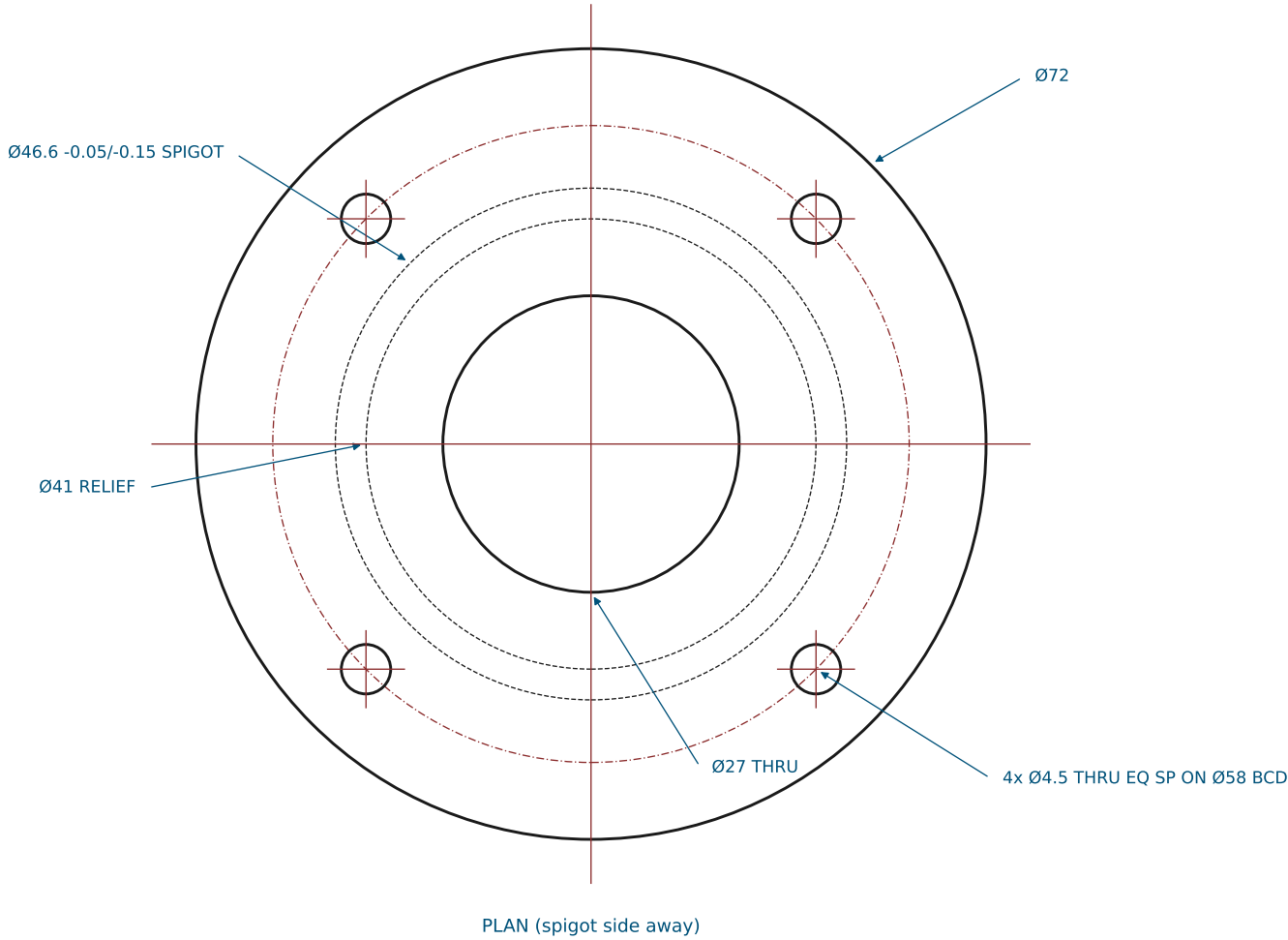


SHEET 1: BEARING RETAINER CAP (BRG-CAP)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

- 1. One cap above AND one below each 7005 bearing bore; the pair bolts together THROUGH the plate with 4x M4x30 SHCS + nyloc (plate holes 4.5 on the same BCD).
- 2. The 46.6 spigot registers in the 47.0 plate bore. Its 41.0-46.6 face ring presses the bearing OUTER race only; the bore is relieved inboard of 41.0 to clear the rotating inner race.
- 3. Build one cap spigot-UP and one spigot-DOWN per bore (same part, flipped).
- 4. OD is 72 so the bolt holes keep a full 1.5d rim (edge distance 4.75).

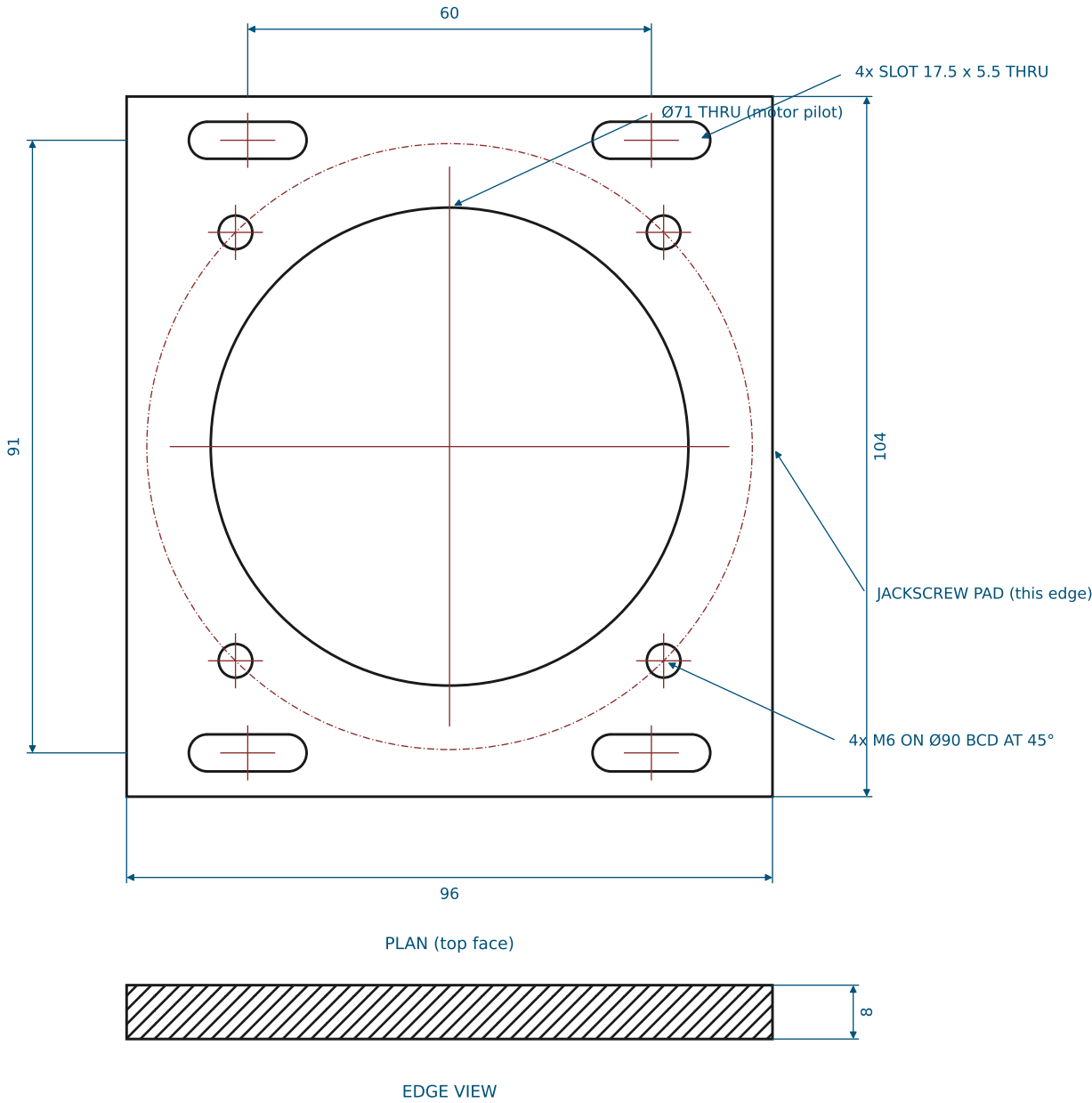


5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: BEARING RETAINER CAP	DWG NO: BCR-01 REV A
MATERIAL: PA12 MJF / 6061-T6	QTY: 8
SCALE 1.5:1 UNITS mm THIRD ANGLE	SHEET 1 OF 6 2026-07-19

SHEET 2: SLIDING MOTOR CARRIAGE (CARRIAGE)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

- 1. Motor (A6M80, 80 sq flange) mounts from BELOW: 4x M6x16 SHCS up through the motor flange into the four M6 positions. PRINTED part: M6 brass heat-set inserts. 6061: tap M6.
- 2. The four slots take M5x16 SHCS + washer down into the cradle frame bosses directly beneath; slot length gives ±6 mm of tension travel.
- 3. The +X edge (toward the shoulder) is the jackscrew pad — the M6 jackscrew tip bears mid-edge.
- 4. Same part both sides (symmetric).

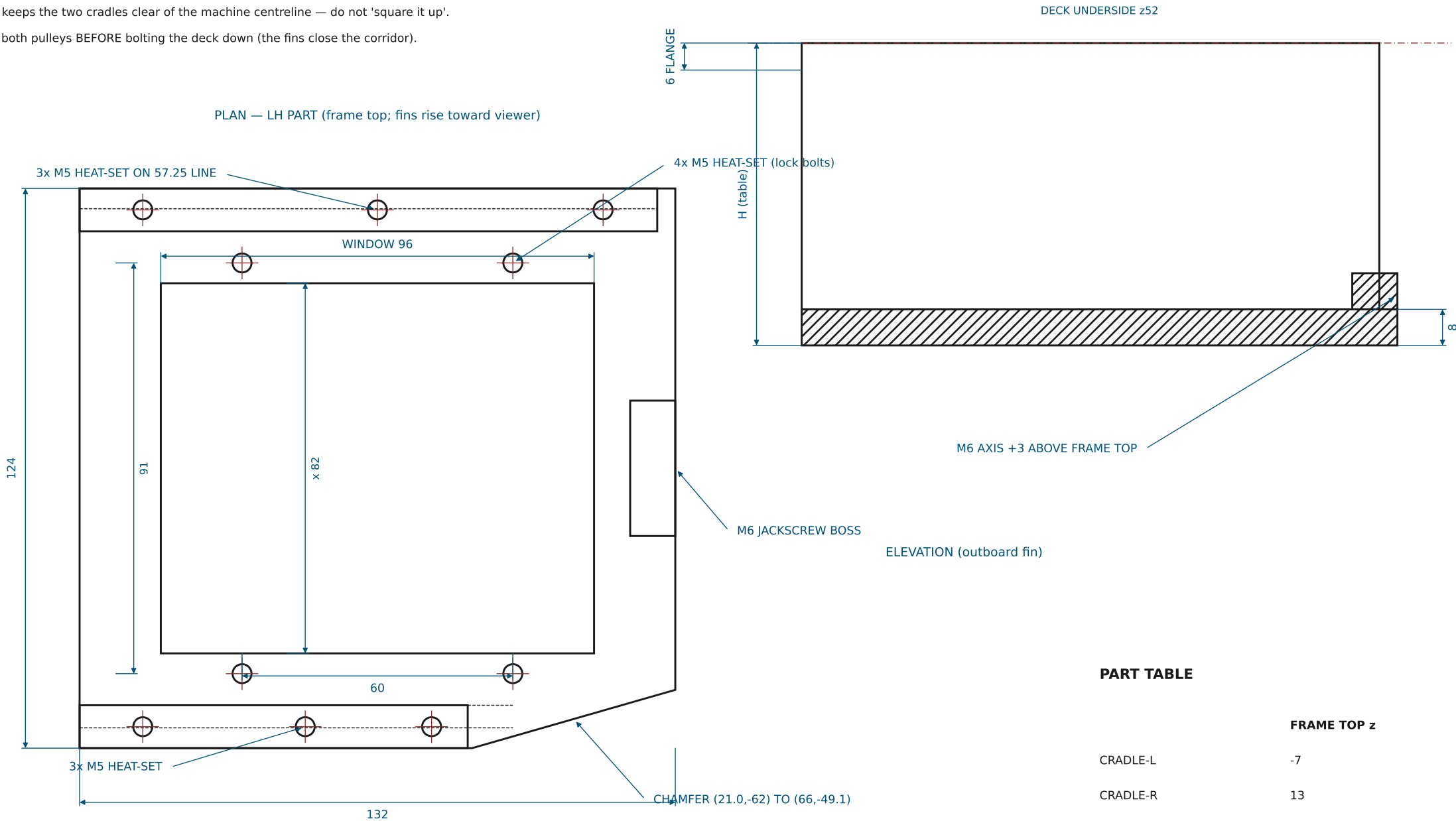


5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: SLIDING MOTOR CARRIAGE	DWG NO: BCR-02 REV A
MATERIAL: PA12 MJF / 6061-T6	QTY: 2
SCALE 1:1 UNITS mm THIRD ANGLE	SHEET 2 OF 6 2026-07-19

SHEET 3: SLIDER CRADLE (LH SHOWN, RH = MIRROR) (CRADLE-L / CRADLE-R)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

1. ONE printed part per motor = window frame + two shear fins + jack block. The RH part is the exact MIRROR of the LH drawn here; fin/wall heights differ per the table (belt planes stagger).
2. Fin top flanges bolt UP into the deck underside: 3x M5 heat-set per fin (deck holes 5.2 THRU, M5x16 SHCS from the deck top).
3. The four M5 positions in the frame border take the carriage lock bolts — fit heat-set inserts.
4. M6 heat-set (or tapped) jackscrew boss in the block; jam nut on the screw against the block face.
5. The chamfered corner keeps the two cradles clear of the machine centreline — do not 'square it up'.
6. Install the belt around both pulleys BEFORE bolting the deck down (the fins close the corridor).



PART TABLE

	FRAME TOP z	WALL H (frame btm→deck)	MIRROR
CRADLE-L	-7	67	as drawn
CRADLE-R	13	47	mirrored

5-BAR BUNG-COVER ROBOT — BASE HARDWARE

TITLE: SLIDER CRADLE (LH SHOWN, RH = MIRROR)	DWG NO: BCR-03 REV A
MATERIAL: PA12 MJF / PETG-CF (print flat, frame down)	QTY: 1 + 1 (mirror)
SCALE 1:2 / 1:1 UNITS mm THIRD ANGLE	SHEET 3 OF 6 2026-07-19

SHEET 4: BOTTOM DECK PLATE (DECK)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

1. Machine per the HOLE TABLE (coordinates from the plate origin = mid-span between the two shoulder bores; +Y toward the arms). All holes THRU.
2. The two 47.0 bores carry the 7005 bearings (transition fit: bore 47.0 +0.025/0).
3. Corner radii R8. Outline vertices in the VERTEX TABLE.
4. Plate must stay within a 300 x 300 print bed (current 294 x 240).

HOLE TABLE

ID	X	Y	Ø	SPEC
B1	40.00	0.00	47.0	7005 bearing bore, thru
B2	-40.00	0.00	47.0	7005 bearing bore, thru
C1	60.51	20.51	4.5	M4 cap bolt, thru
C2	19.49	20.51	4.5	M4 cap bolt, thru
C3	19.49	-20.51	4.5	M4 cap bolt, thru
C4	60.51	-20.51	4.5	M4 cap bolt, thru
C5	-19.49	20.51	4.5	M4 cap bolt, thru
C6	-60.51	20.51	4.5	M4 cap bolt, thru
C7	-60.51	-20.51	4.5	M4 cap bolt, thru
C8	-19.49	-20.51	4.5	M4 cap bolt, thru
S1	64.00	36.00	5.2	M5 standoff bolt, thru
S2	64.00	-36.00	5.2	M5 standoff bolt, thru
S3	-64.00	36.00	5.2	M5 standoff bolt, thru
S4	-64.00	-36.00	5.2	M5 standoff bolt, thru
F1	136.25	-127.95	5.2	M5 fin bolt, thru
F2	121.91	-77.96	5.2	M5 fin bolt, thru
F3	108.13	-29.90	5.2	M5 fin bolt, thru
F4	26.18	-159.51	5.2	M5 fin bolt, thru
F5	16.26	-124.90	5.2	M5 fin bolt, thru
F6	8.54	-97.99	5.2	M5 fin bolt, thru
F7	-136.25	-127.95	5.2	M5 fin bolt, thru
F8	-121.91	-77.96	5.2	M5 fin bolt, thru
F9	-108.13	-29.90	5.2	M5 fin bolt, thru
F10	-26.18	-159.51	5.2	M5 fin bolt, thru
F11	-16.26	-124.90	5.2	M5 fin bolt, thru
F12	-8.54	-97.99	5.2	M5 fin bolt, thru

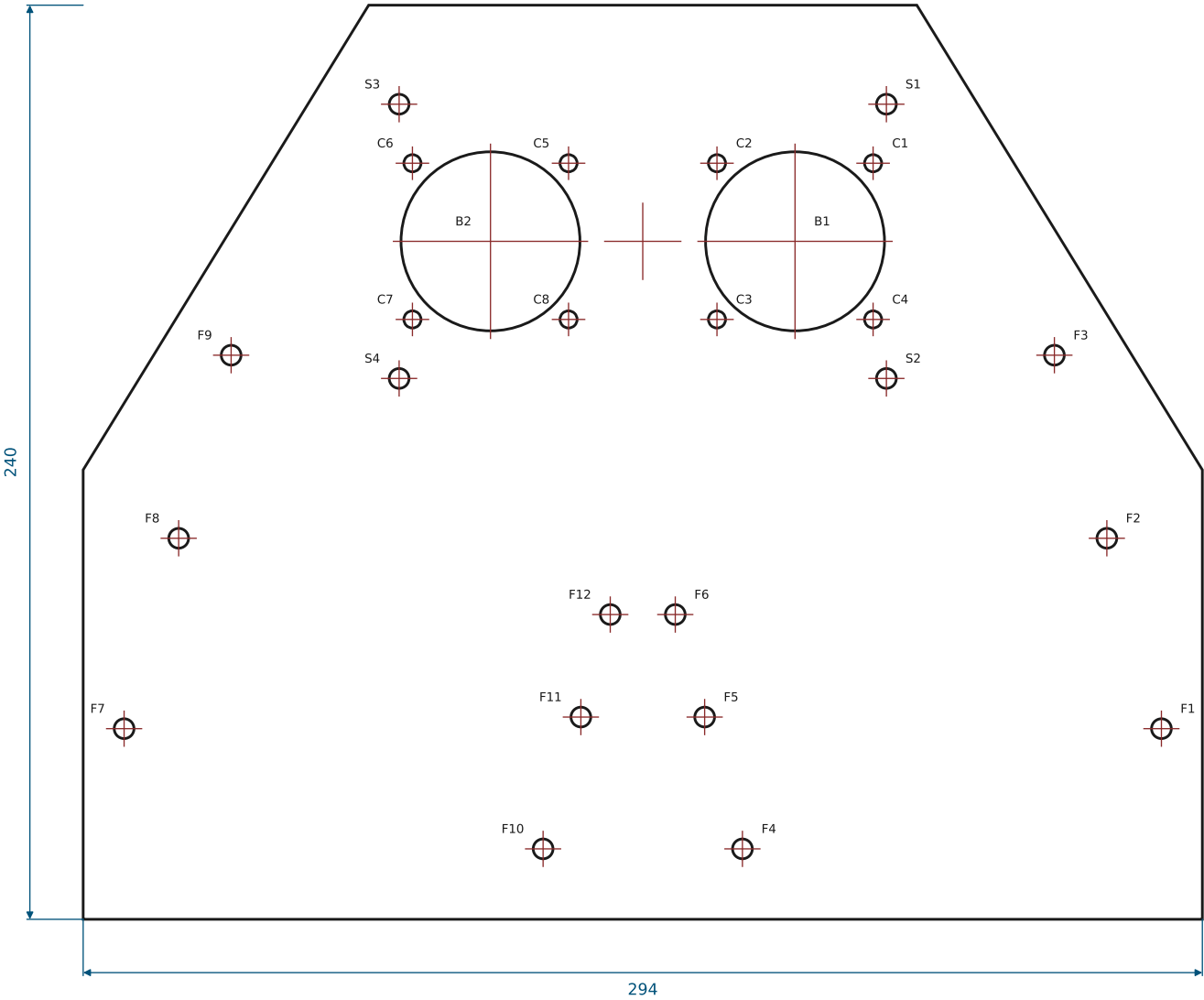
VERTEX TABLE (outline, R8 corners)

V1 (-72,62)	V2 (72,62)	V3 (147,-60)
V4 (147,-178)	V5 (-147,-178)	V6 (-147,-60)

5-BAR BUNG-COVER ROBOT — BASE HARDWARE

TITLE: BOTTOM DECK PLATE	DWG NO: BCR-04 REV A
MATERIAL: PA12 MJF / 6061-T6 (12 THK)	QTY: 1
SCALE 1:2 UNITS mm THIRD ANGLE	SHEET 4 OF 6 2026-07-19

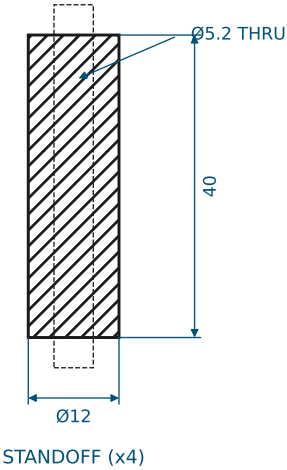
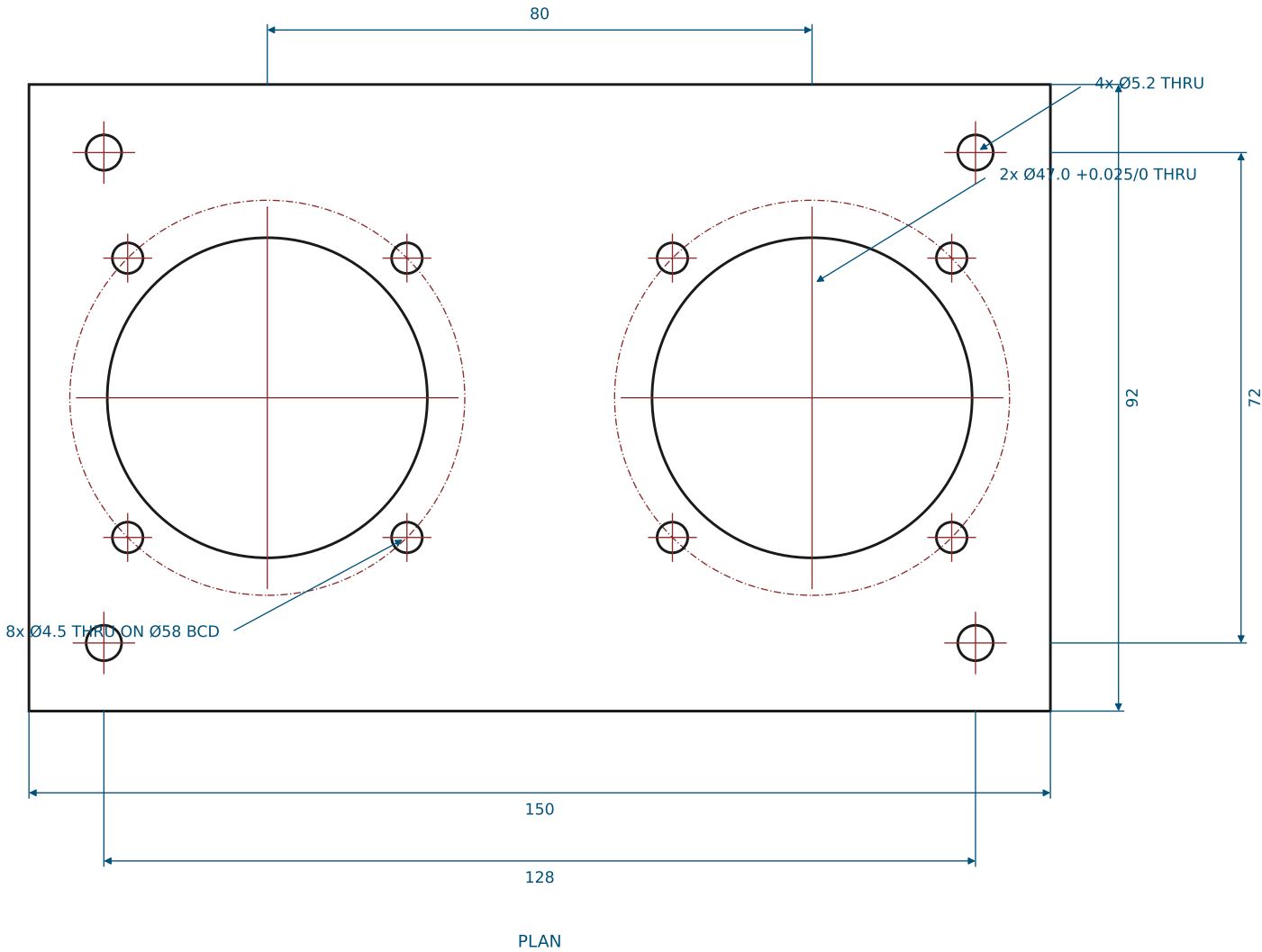
PLAN — deck top (origin at mid-span of shoulder bores)



SHEET 5: TOP PLATE + STANDOFFS (TOP-PLATE)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

- 1. Same bearing bore + cap-bolt pattern as the deck (47.0 bores, 4x 4.5 on Ø58 BCD per bore).
- 2. Standoff bolt holes moved OUT to (±64, ±36) so the Ø72 caps clear the standoff bosses (they collided with the old Ø62 caps at ±60,±32).
- 3. 4x standoffs Ø12 x 40 (M5 clearance 5.2 bore), deck top to plate bottom.

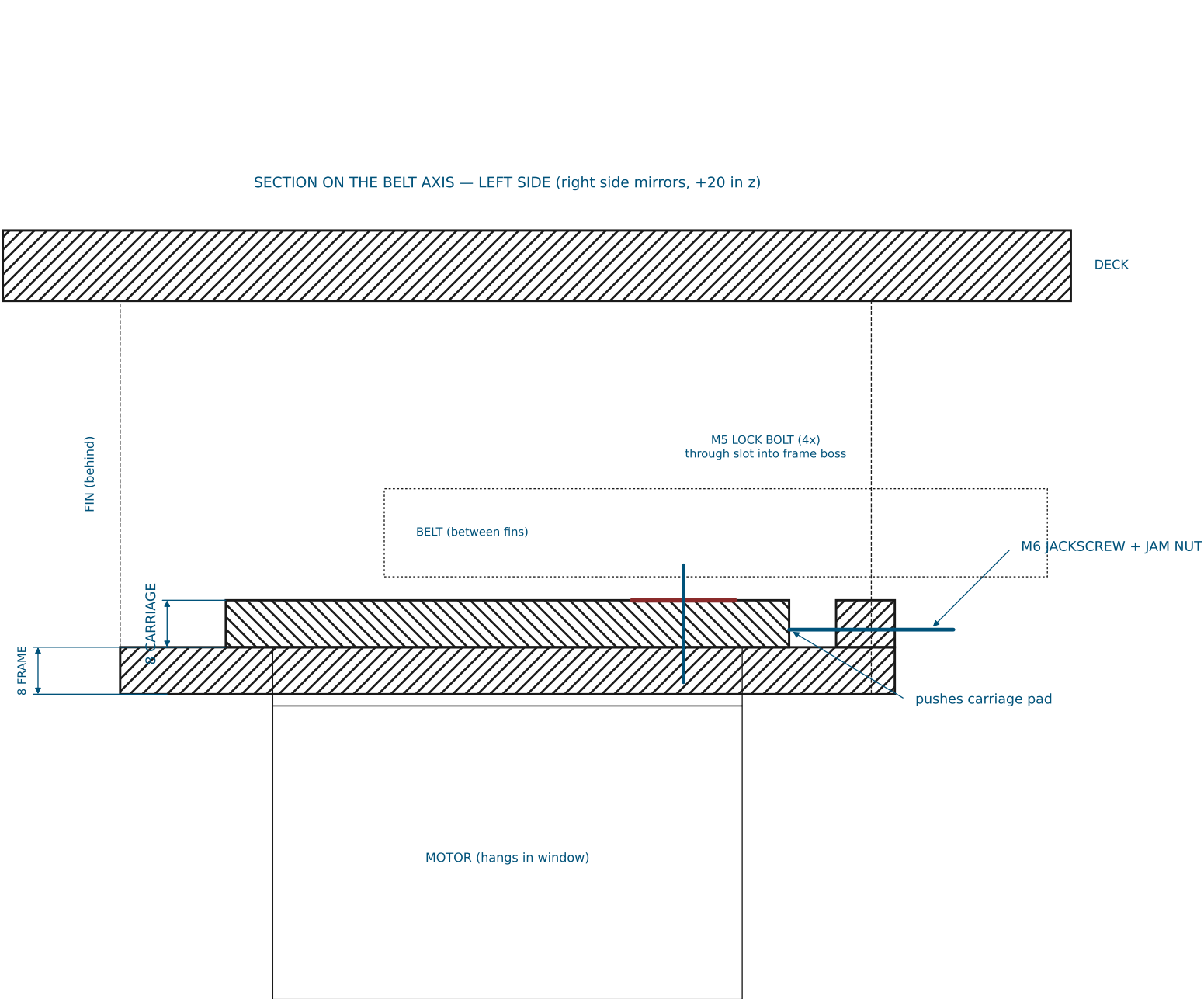


5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: TOP PLATE + STANDOFFS	DWG NO: BCR-05 REV A
MATERIAL: PA12 MJF / 6061-T6 (10 THK)	QTY: 1 (+4 standoffs)
SCALE 1:1 UNITS mm THIRD ANGLE	SHEET 5 OF 6 2026-07-19

SHEET 6: TENSIONER ASSEMBLY + HARDWARE (ASSY-TENSIONER)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

1. TENSION PROCEDURE: loosen the 4x M5 lock bolts 1/2 turn -> turn the M6 jackscrew CW to push the carriage away from the shoulder (belt tightens) -> torque the lock bolts (M5: 4 N·m in inserts) -> set the M6 jam nut against the block.
2. Travel available ±6 mm. Belt: 450-5M-15 (24T -> 72T, C = 97.5 nominal).
3. ASSEMBLY ORDER: pulleys + belt on motor/shaft FIRST, then lower the deck onto fins + standoffs and bolt down (the fin walls close the belt corridor once the deck is on).
4. The belt passes BETWEEN the two fins, above the frame and below the deck — nothing crosses it.



HARDWARE (per machine)

- 8x M4 x 30 SHCS + nyloc + washers
bearing cap pairs, thru plate
- 8x M5 x 16 SHCS + washer
carriage lock bolts (4 per side)
- 12x M5 x 16 SHCS
fin flanges -> deck underside
- 8x M5 x 16 SHCS
deck/top-plate standoff bolts
- 8x M6 x 16 SHCS
motor flange -> carriage (from below)
- 2x M6 x 40 jackscrew + jam nut
tensioner
- 2x M25x1.5 shaft locknut (KM5)
bearing preload
- M5/M6 brass heat-set inserts
all printed bosses

5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: TENSIONER ASSEMBLY + HARDWARE	DWG NO: BCR-06 REV A
MATERIAL: -	QTY: 2 (mirrored)
SCALE 1:1 section UNITS mm THIRD ANGLE	SHEET 6 OF 6 2026-07-19